

Week 7: React.js Basics

1. Introduction to React

React.js is a popular JavaScript library used for building user interfaces, especially single-page applications. It allows developers to create reusable UI components and manage dynamic content efficiently. React improves performance using the virtual DOM.

2. Components, Props, State

Components are reusable building blocks in React. Props are used to pass data from parent to child components. State is used to manage dynamic data inside a component. Understanding these concepts is essential for creating interactive React applications.

3. JSX, Event Handling in React

JSX is a syntax extension that allows writing HTML-like code inside JavaScript. It makes UI development easier and more readable. Event handling in React is similar to JavaScript but uses camelCase syntax. It helps respond to user interactions like clicks and form submissions.

4. useEffect Hook, Conditional Rendering

The `useEffect` hook is used to handle side effects such as API calls, timers, or updating the DOM. Conditional rendering allows displaying UI elements based on conditions. These features make React applications more dynamic and efficient.

Conclusion

This week focused on React.js basics. By learning components, props, state, JSX, event handling, `useEffect`, and conditional rendering, a strong foundation for frontend application development using React was established.